Sugarcane Procurement Management Using Blockchain

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***Abstract*—One of the problems with the existing sugar cane procurement program is that there is often no way to determine if the information in records has been changed or falsified through collusion between farmers and millers. In this project, we propose a blockchain-based sugarcane procurement system which will provide a reliable, open and tamper-proof way to conduct transactions. In this blockchain-based sugarcane procurement system, the farmer begins the transaction by registering the sugar cane in lots by date harvested, amount harvested, and the location where it was harvested. Then a smart contract will validate the information submitted to it and give each lot an identifying number. After these steps are completed, the information is forwarded to the miller to update and approve in real-time that the information they received from the farmer is accurate based on their harvesting and getting an approved sale of the sugar cane. When a sale is made of the sugar cane, the farmer will update the information on their sale to the miller. Once this transaction has been completed, the historical sale records are now recorded in the public blockchain which means they cannot be changed. Because of this, the chance of fraud or tampering with the recorded sale information of sugar cane will no longer exist, therefore enhancing the sugarcane procurement process's efficiency and removing the potential for conflict between the farmer and the miller. With a blockchain-based sugar cane procurement system being put into place, the farmer's actions will now have a high level of transparency through the procurement system and receive the recognition of their contribution to the procurement process in a more timely manner. This new blockchain-based sugarcane procurement system should dramatically improve on the way the sugarcane is procured by the miller.**

***Index Terms*—Blockchain, Sugarcane Procurement, Smart Contracts, Decentralization, Immutable Ledger, Transparency.**

I. Introduction

A story of complete faith in a transparent procurement system for sugarcane is provided by the creation of a blockchain-based sugarcane procurement system. Normally sugarcane is bought from a farmer, transported to the transporters' locations, and then sent to the sugar mills. The process has many records at every stage; however, whether this information can provide all of the data that is recorded with confidence, by all participants in each link of the supply chain, is the critical challenge. Fairness issues exist in many facets of the procurement system, including weight, quality, time, and payment. It is here that the use of blockchain technology becomes the ideal solution [5]. To build a procurement system based on blockchains, an understanding of what a blockchain is (an overall concept) and how it can be used in many different applications is required. One key element in achieving this is accepting that a procurement system needs to not only store data but also be a collaborative platform where all stakeholders can work together without ambiguity, for example via the blockchain as a platform with online tools, is where sugarcane procurement will be, and is how to start building the system [1]. As the use of blockchains continues to advance, it is essential that blockchains are used in conjunction with additional technologies such as the Internet of Things (IoT). In a real life example such as procurement, data is not received from a single source (e.g. one level of input), but is collected from multiple sources (e.g. multiple types of input) – in this case, data is received from sources such as: sensors, scales/weighing machines, transportation and confirmation of delivery. Using IoT and other forms of technology can help to combine these multiple sources of data and work together (in a collaborative manner) without having a single source. In addition to connecting individual sources of data, automated contracts can be used, e.g. to perform automated functions such as confirming receipt of goods and processing payments. In the example of sugarcane procurement, this will serve to connect together all aspects of the procurement process, both in the field and in the mill [2]. From a broader perspective, blockchain technology isn't only about cryptocurrencies anymore; it now also covers additional applications like supply chain management. This is crucial, as it shows that the issues associated with sourcing sugarcane are not isolated; they are a part of larger global supply chain inefficiencies. However, the adoption of blockchain in supply chain management does have challenges. The system must balance user-friendliness and transparency [3]. Considering the technical aspects of the blockchain, it is evident that the blockchain's structure is extremely important for the system to work well. Factors like blockchain structure and overall system architecture influence the system's overall efficiency. In the case of sugarcane sourcing, the structure of the blockchain will have a crucial effect on how smoothly the sourcing system operates [4]. When examining agriculture specifically, one can see that there is great potential for using blockchain to solve some of the issues faced by the industry. Corporately, with regard to agriculture's supply chain, there are problems involving inefficiently operating entities within the supply

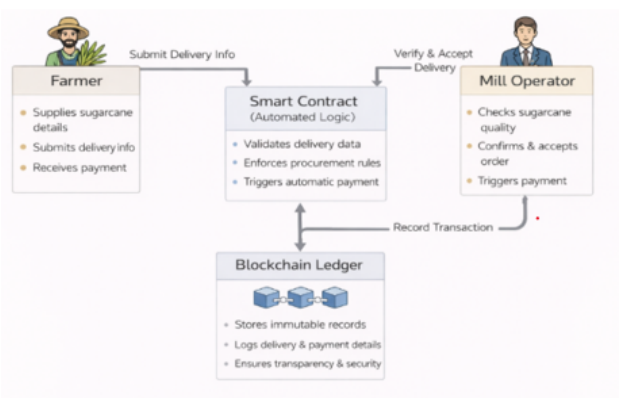
chain (e.g., lack of communication between farmers and food processors) due to factors such as communication breakdown between supply chains and uncoordinated suppliers. Fortunately, blockchain technology creates new opportunities for improved traceability and collaboration within the agriculture supply chain. This is particularly applicable to the procurement of sugarcane, enabling stakeholders to more effectively track the sugarcane crop from the grower to the processor while providing improved communication among all parties involved [6]. Notably, smart contracts will be able to revolutionise the procurement process in respect of the sugarcane Supply Chain; this is achieved by removing some of the processes currently required to approve the procurement of sugarcane through traditional means. Smart contracts enable all parties in the supply chain to work together only when it can be verified that the requirements of the supply chain have been met so as to facilitate payment to farmers when sugarcane is delivered to the processor without invoking any administrative processes or requiring any physical documents [7]. Furthermore, the development of blockchain technology has changed the way businesses perceive trust between parties. Traditionally, trust was built on the basis of third parties, written documentation or records, and repeated validation. By using a distributed ledger (like Blockchain) instead of relying solely on these methods for sugarcane supply chain procurement, sugarcane suppliers and processors can now trust each other's transactions without requiring a third party. With this new level of digital trust, sugarcane suppliers and processors can eliminate unnecessary middlemen or redundant processes and build a much more efficient logistics supply chain [8]. Real-world examples further illustrate how all products can be traced throughout their whole life-cycle. And although sugarcane is not a food product sold at retail, it is important to realize that each step in sugarcane's life cycle (from farming to finishing) can be tracked back to the original source. As such, all sugarcane procurement will be done with complete transparency [9]. From a more macro perspective, blockchain technology represents a fundamental shift in the way that transactions (or collaborative activities) are organized. This is not merely a shift in technology, it is also a shift in the way people establish trust in business operations. The procurement of sugarcane, for example, has been based on fragmented/paper processes – moving instead to an integrated/digital way of doing business benefits everyone in the supply chain [10]. In order for supply chains to use blockchain technology effectively, we must understand the operational challenges associated with the implementation of such a system. We must develop the sugarcane procurement procedure with a clear knowledge of those challenges to ensure successful implementation. In addition to aligning blockchain technologies with the real-world requirements/constraints, we will also be able to experience the advantages of using blockchain technologies [11]. One key aspect of procurement systems is the ability to trace the entire journey of the sugarcane; this is critical for guaranteeing quality. By using block-chain technology to achieve traceability in procurement systems, all transactions and events within the procurement system have an unbroken record. The ability to quickly verify information, quickly settle disputes and build trust between consumers and suppliers through traceability is greatly enhanced with the introduction of block-chain technology [12]. In addition to the improvement of traceability, the enhancement of block-chain frameworks has proven how procurement systems can also be improved by block-chain technology with regard to the verification and origin for quality standard. The use of structured workflows and validations in procurement systems also provides improvements to procurement systems. In the case of sugarcane procurements, the improvements to the procurement systems allows for better quality control at the various quality check and validation points [13]. Sharing of Data is another key consideration when determining how effective supply chains are. An example of this can be found when utilizing procurement systems; they typically require multiple users to share data at different stages along their respective paths. Data stored on the blockchain, however, can be accessed by all parties involved (as long as they have permissions). As a result, this type of data sharing will facilitate the movement of goods and services, which will allow for more efficient communication amongst users [14]. Assessment of performance levels for all newly implemented information-based technologies is another major consideration of new technology. In regards to assessing whether a blockchain based solution is beneficial or not, procurement processes will need to demonstrate increased efficiency, transparency, and reliability as an outcome [15]. When considering the overall design of information based systems, some design factors need to be taken into account at their initial implementation level (e.g. scalability, privacy, and compatibility). For example, in order for a sugar cane procurement process to be practical, it may require multiple transactions while still keeping private information about both buyers sellers secure [16]. Sustainability also plays an extremely important part in this. A big way that blockchain can assist in building a more sustainable procurement system, is by creating a transparent and efficient procurement process. This transparency and efficiency will be seen with sugarcane procurement. Security and trust are also an integral part of the blockchain [17]. The blockchain has both a decentralised and immutable quality, meaning that once information has been placed into the blockchain, it cannot be changed and can therefore be relied on with confidence; This greatly reduces the risk of fraudulent procurement and maximises the reliability of the information collected [18]. By integrating blockchain technology with other Digital Technologies, the Overall impact of all technologies is enhanced since Blockchain is now part of a larger Digital Systems Together When Blockchain Works in Conjunction with Other Technologies Such As Sensors, Data Analysis, And Automation, It Enhances The Accuracy Of The Data Being Collected, Monitored, And Utilized In The Purchase Of Sugarcane [19]. Recent trends show that agri-food supply chains have changed when it comes to resilience, sustainability, and traceability; Blockchain is seen as an important element of that change. As a result, the systems

that are being developed for procuring Sugarcane will be able to not only address current issues but also will be able to meet future needs which will, in turn, create greater efficiencies over time [20].

II. Research Methodology

The objective of this dissertation is to develop a blockchain-based application to address challenges associated with sugarcane procurement. This research will use a combination of literature review and existing system analysis (problem definition and requirements) to analyze current challenges in the sugarcane procurement process. The next step will be to design a fully decentralized, blockchain-based application utilising smart contracts and stakeholder-defined roles. This will create a prototype on the Ethereum platform, which will then be tested against several case study examples. Finally, the system will be validated against real-world examples to demonstrate its effectiveness, transparency, and scalability.

III. Transferring of Assets



IV. Blockchain Architecture

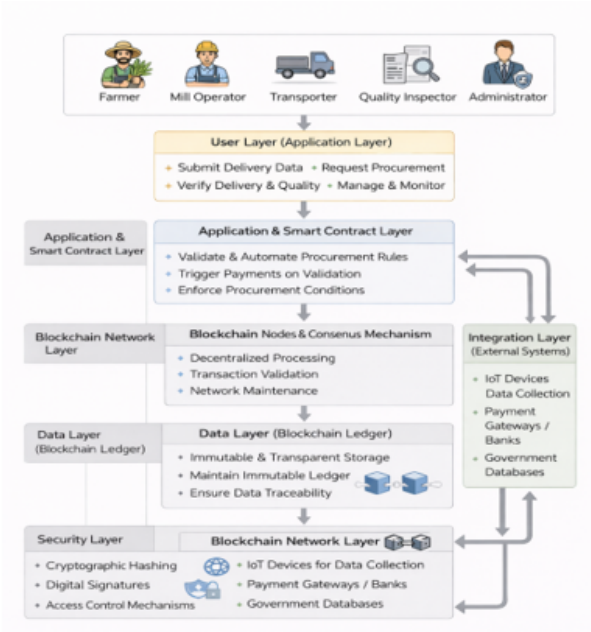


Fig. 1. Blockchain Architecture

The Sugarcane Procurement Management System utilizes a multi-layer blockchain structure to establish transparency, automate transactions, and securely manage records.

- **User Layer:** This layer contains all the system participants. It is made up of farmers, millers, truckers, quality inspectors, and administrators. Users utilize the system for submitting delivery records, requesting procurement, verifying product quality, and performing all financial transactions.
- **Application and Smart Contract Layer:** This layer contains the application logic of the Sugarcane Procurement Management System. It consists of smart contracts that verify delivery-related information, execute procurement requests, and create payment requests, all of which require the satisfaction of contractual conditions to proceed.
- **Blockchain Network Layer:** The blockchain network layer is made up of many different distributed nodes that verify each transaction through consensus. This allows for decentralized processing of transactions, assurance of data integrity, and reliability of the system.
- **Data Layer:** Every transaction conducted in the system, including deliveries, quality verification, and payment transactions will also be recorded in the data layer of the system. The information recorded in this layer cannot be altered or changed by any participant.
- **Security Layer:** The security layer consists of various security features including hash functions, digital signatures, and access controls, that provide for the secure operation of the system as a whole and the secure disposition of all transferred data.

V. List of Stakeholders

- **Farmer:** Records crop information, submits sugarcane delivery requests, enters quality and quantity data, and receives payment after verification of data.
- **Mill Operator:** Transforms orders for sugarcane and checks the quality/weight of the sugarcane in the system, approves/denies delivery of sugarcane and pays farmers using the Blockchain.
- **Transporter / Logistics Provider:** Arranges transportation of sugarcane from farms to mill, maintains an updated status about the delivery of goods and ensures timely delivery of goods with the appropriate level of security while making sure that the goods are handled properly during shipping to their destination.
- **Smart Contract (System Component):** Record crop information, create sugar cane delivery requests, enter quality/quantity data, and transfer payment for successful verification.
- **Blockchain Ledger:** Request to purchase, verify quality/weight of sugar cane, approve/reject delivery, and initiate payment to farmers via blockchain.
- **Quality Inspector:** Evaluates the quality of sugar cane using pre-defined parameters and updates the verification results in the system.
- **Bank / Payment Gateway:** Provides the means for digital payments between a mill operator and a farmer, using blockchain for secure payment transactions.
- **Government Authority / Regulatory Body:** Modernizes purchases; assures compliance with agricultural regulations; and resolves any disputes that may arise between stakeholders.
- **System Administrator:** Maintains the blockchain network by securing individual nodes, monitoring and maintaining system performance, and controlling access to all users.

VI. Conclusion

The propose Blockchain-Based Sugarcane Procurement Management System was designed to address issues that have plagued the traditional procurement process, such as lack of trust by not allowing visibility into transaction verification, delayed payments to farmers, inconsistency in storehouse reports from the mill, manipulation of records, and lack of visibility into the location of sugarcane in the procurement process. The goal of the new procurement management system will provide for each transaction performed in the procurement process, whether it be in the receipt of crops by the mill from the farmer or in the settlement of payment to the farmer, to be completed securely, immutably, and transparently using the features enabled through the use of blockchain technology.

On consummation of a contract using a smart contract within the new system, the smart contract will ensure that the quality of sugarcane delivered by each farmer is evaluated and subsequent payment to the farmer is settled timely without having the need for an intermediary. This will also provide a greater level of trust between the mill and the farmer. In addition, the decentralized structure of the blockchain technology being used in the procurement process will provide assurance that the records maintained within the new system have been maintained with integrity, thereby providing for all stakeholder(s) access to accurate, reliable and current status for the sugarcane being procured through the new procurement management system. In summary, the procurement management system described herein has the capacity to change the conventional how sugarcane is procured by improving the efficiency, transparency, and integrity of the process by which sugarcane is procured.

References

[1] P. Raj, G. C. Deka, and R. Buyya, *Blockchain Technology: Platforms, Tools and Use Cases*, vol. 111, Amsterdam, Netherlands: Elsevier, 2017.

[2] K. Christidis and M. Devetsikiotis, "Blockchains and Smart Contracts for the Internet of Things," *IEEE Access*, vol. 4, pp. 2292–2303, 2016.

[3] F. Casino, T. K. Dasaklis, and C. Patsakis, "A Systematic Literature Review of Blockchain-Based Applications: Current Status, Classification and Open Issues," *Telematics and Informatics*, vol. 36, pp. 55–81, 2019.

[4] Z. Zheng, S. Xie, H. Dai, X. Chen, and H. Wang, "An Overview of Blockchain Technology: Architecture, Consensus, and Future Trends," *Proc. IEEE International Congress on Big Data*, pp. 557–564, 2017.

[5] N. Kshetri, "Blockchain's Roles in Meeting Key Supply Chain Management Objectives," *International Journal of Information Management*, vol. 39, pp. 80–89, 2018.

[6] A. Rejeb, K. Rejeb, S. Simske, and A. Ghorbel, "Blockchain Technology in the Agri-Food Industry: A Systematic Literature Review and Research Agenda," *Logistics*, vol. 5, no. 1, pp. 1–26, 2021.

[7] R. Sharma and P. Kumar, "Smart Contracts and Blockchain for Transparent Agricultural Supply Chains," *Proc. International Conference on Smart Electronics and Communication*, pp. 112–118, 2020.

[8] W. Mougayar, *The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology*, Hoboken, NJ, USA: Wiley, 2016.

[9] IBM Corporation, "Blockchain in Agriculture Use Cases," *IBM Blockchain Documentation*, Online.

[10] D. Tapscott and A. Tapscott, *Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies Is Changing the World*, New York, NY, USA: Portfolio, 2016.

[11] P. Dutta, T.-C. Choi, S. Somani, and R. Butala, "Blockchain Technology in Supply Chain Operations: Applications, Challenges and Research Opportunities," *Transportation Research Part E*, vol. 142, 2020.

[12] G. Mirabelli and I. S. Lami, "Blockchain and Agricultural Supply Chain Traceability: Research Trends and Future Challenges," *Procedia Manufacturing*, vol. 42, pp. 414–421, 2020.

[13] F. Mokgomola, "Blockchain and Smart Contracts Based Agricultural Supply Chain," *Journal of Agricultural Informatics*, vol. 15, no. 1, pp. 34–45, 2024.

[14] N. N. Abdullah et al., "Blockchain Adoption in the Supply Chain: Emerging Trends and Future Research Directions," *Journal of Supply Chain Management*, vol. 61, no. 1, pp. 1–18, 2025.

[15] A. Kamilaris, A. Fonts, and F. X. Prenafeta-Boldu, "The Rise of Blockchain Technology in Agriculture and Food Supply Chains," *Trends in Food Science and Technology*, vol. 91, pp. 640–652, 2019.

[16] O. Karaduman, "Blockchain-Enabled Supply Chain Management: A Review," *Computers and Industrial Engineering*, vol. 198, 2025.

[17] R. M. Difrancesco, "Blockchain for Sustainable Supply Chain Performance," *Sustainability*, vol. 14, no. 3, pp. 1–19, 2022.

[18] M. Khatun, "Unlocking Blockchain's Potential in Supply Chains," *Journal of Business Logistics*, vol. 46, no. 2, pp. 155–170, 2025.

[19] M. Vasileiou, "Digital Transformation of Food Supply Chains Through Blockchain," *Food Control*, vol. 165, 2025.

[20] R. Silvestri, "Blockchain in Agri-Food Supply Chains: Traceability, Transparency and Sustainability Challenges," *Trends in Food Science and Technology*, vol. 102, pp. 120–134, 2025.